

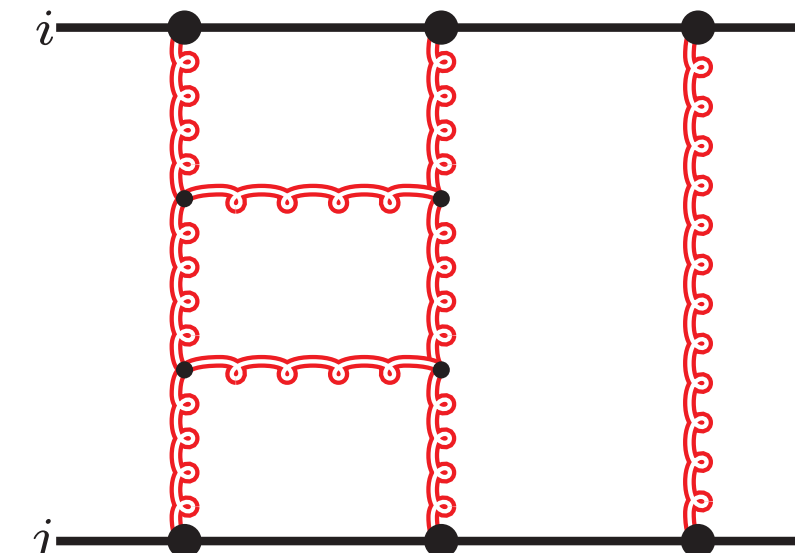
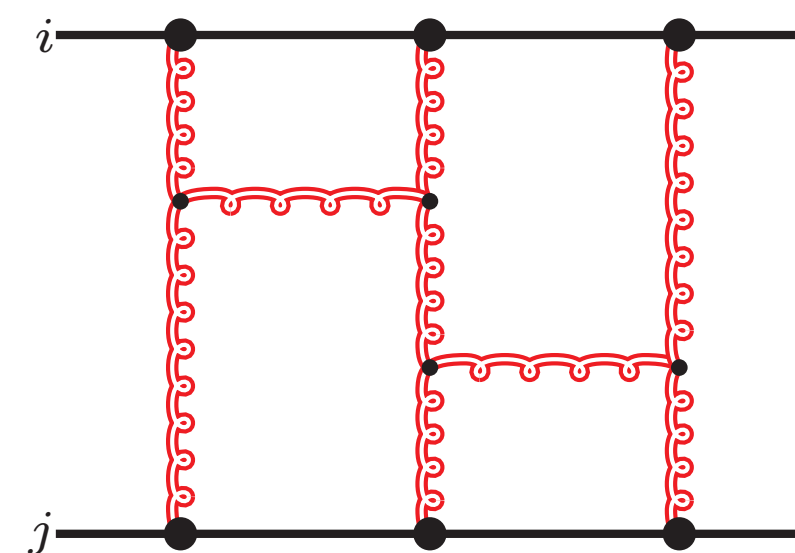
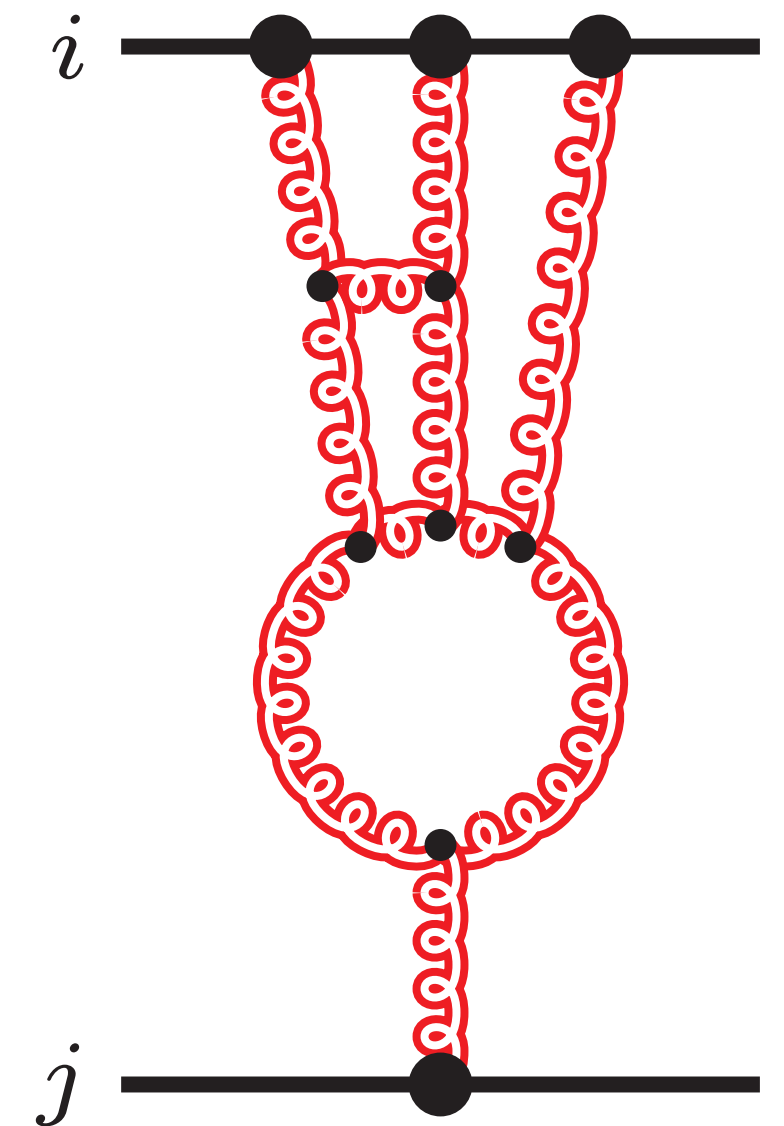
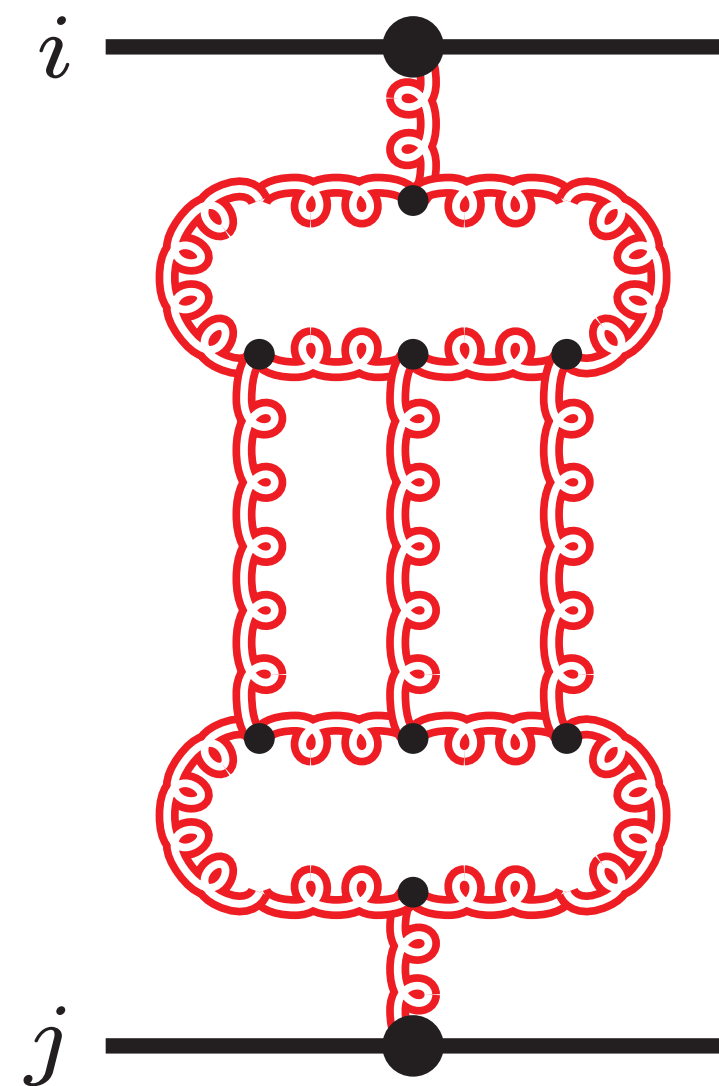
# Disentangling the **Regge** pole and **Regge** cut in perturbative QCD

Calum Milloy

[PRD 103 (2021), 2012.00613] w/ Giulio Falcioni, Einan Gardi and Leonardo Vernazza

[JHEP 03 (2022) 053, 2111.10664] & [PRL 128 (2022) 13, 2112.11098]  
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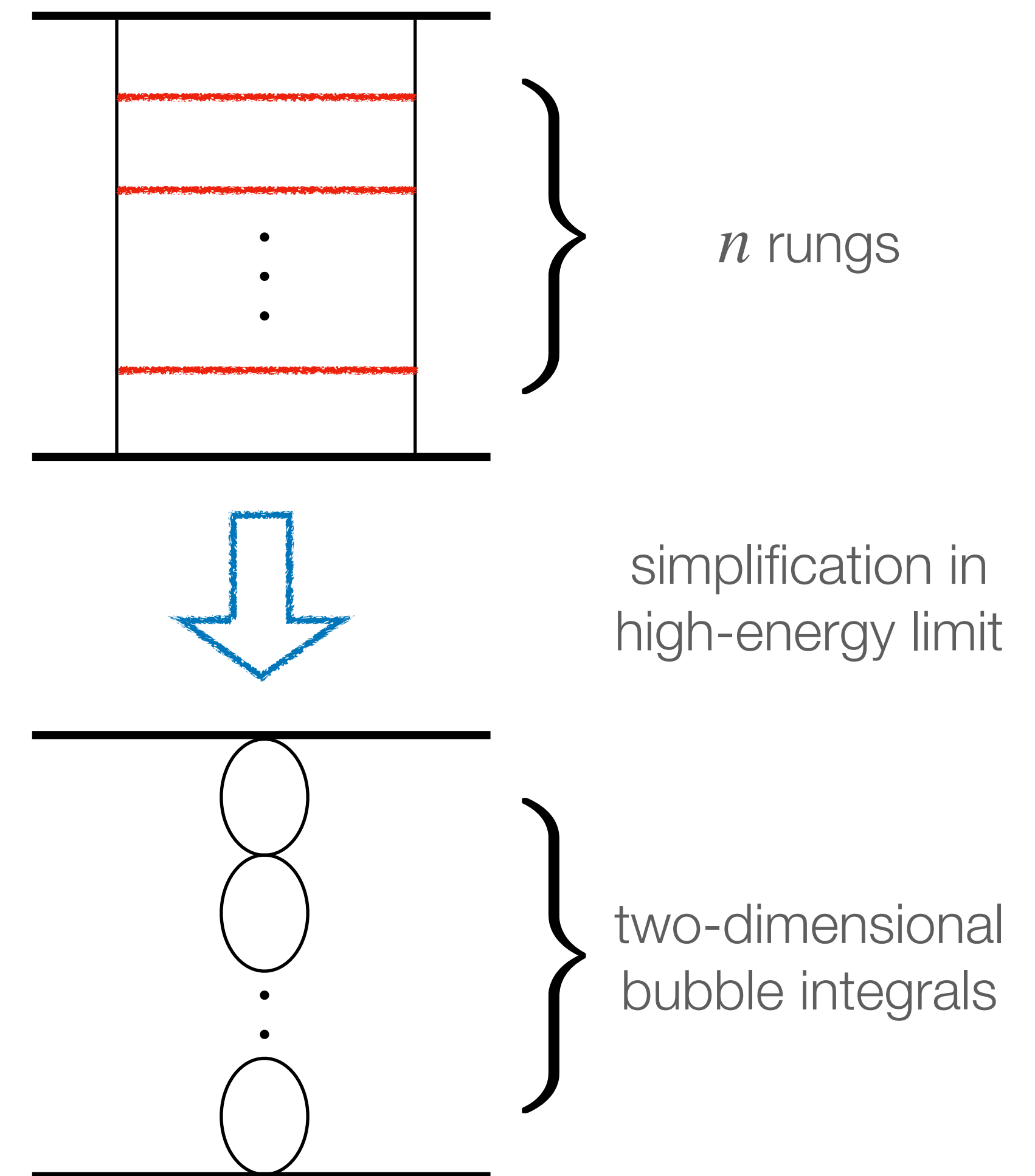
# Contents

- Introduction to the **high-energy limit** of  $2 \rightarrow 2$  scattering amplitudes
  - What are Regge poles and cuts?
  - Leading Logarithm (LL) Approximation
  - Next-to-Leading Logarithms (NLL)
- NNLL
  - Calculation of multi-Reggeon exchanges
  - How to distinguish **pole** and **cut** contributions
- Applications
  - The **three-loop Regge trajectory** in QCD
  - Infrared constraints

# Motivation

Why care about the high-energy limit?

- Interesting **theoretical** question
- Use it as a **toy model** for scattering amplitudes
- Has still rich structure in both **kinematics** and **colour**
- Understand **all-order behaviour** of amplitudes
- Regge **poles** and Regge **cuts** in perturbative QCD
- Provide constraints for **universal** infrared divergences





# 2 → 2 Scattering Amplitudes

- We will consider 2 → 2 scattering amplitudes in **any** massless theory

- $i(p_1, a_1, \lambda_1) + j(p_2, a_2, \lambda_2) \rightarrow j(p_3, a_3, \lambda_3) + i(p_4, a_4, \lambda_4)$

- Described in terms of Mandelstam invariants

- $s = (p_1 + p_2)^2 > 0$        $t = (p_1 - p_4)^2 < 0$        $u = (p_1 - p_3)^2 < 0$

momentum conservation implies  $s + t + u = 0$

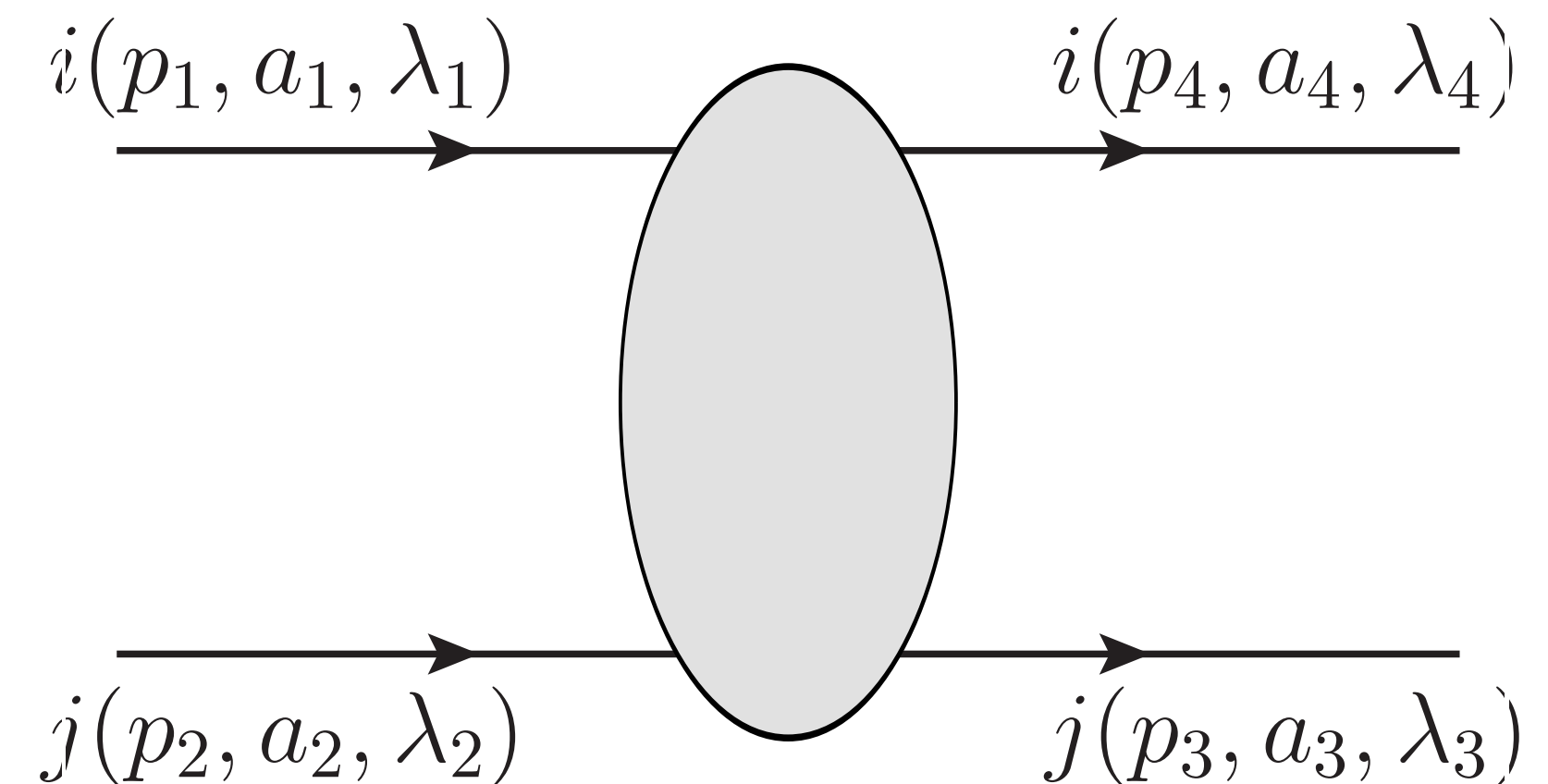
- The **high-energy limit** is defined to be  
centre-of-mass energy much larger than momentum transfer

$$s \gg -t$$

$$s \approx -u$$

amplitudes have crossing symmetry properties under

$$s \leftrightarrow u$$



# What are Regge poles and cuts?

Let us travel back to the 1960s, prior to QCD

[Regge '59, '60; Eden, Landshoff, Olive, Polkinghorne '66; Collins '77]

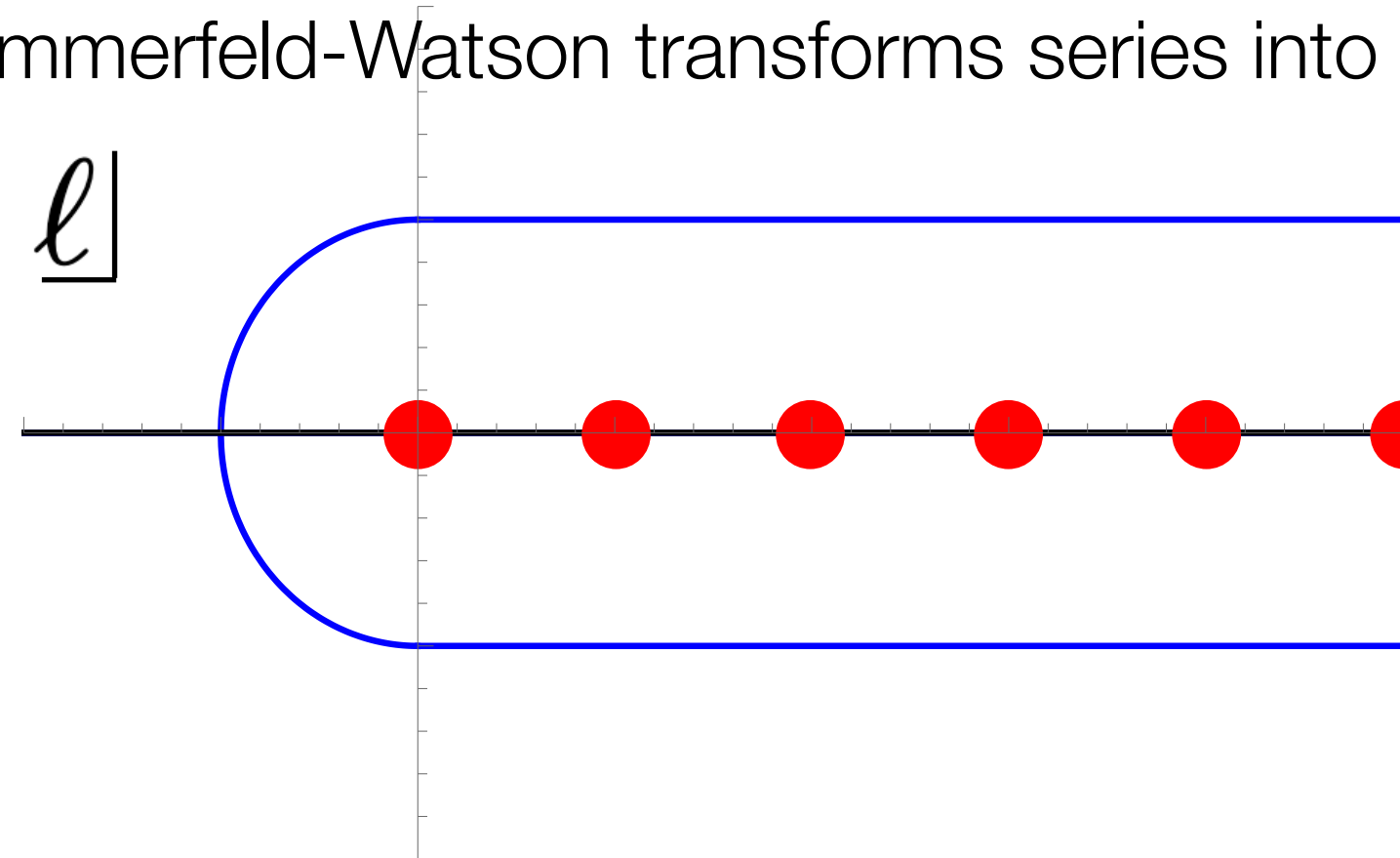
Start with partial wave expansion of the scattering amplitude

$$\mathcal{M} \sim \sum_{\ell=0}^{\infty} (2\ell + 1) A_{\ell}(t) P_{\ell}(s)$$

angular momentum in t-channel

dependence of a state of angular momentum  $\ell$  are the Legendre polynomials

Sommerfeld-Watson transforms series into a counter integral, picking up poles in the **complex angular momentum plane**



$$\mathcal{M} \sim \oint d\ell (2\ell + 1) A_{\ell}(t) P_{\ell}(s) \frac{\pi}{\sin \pi \ell}$$

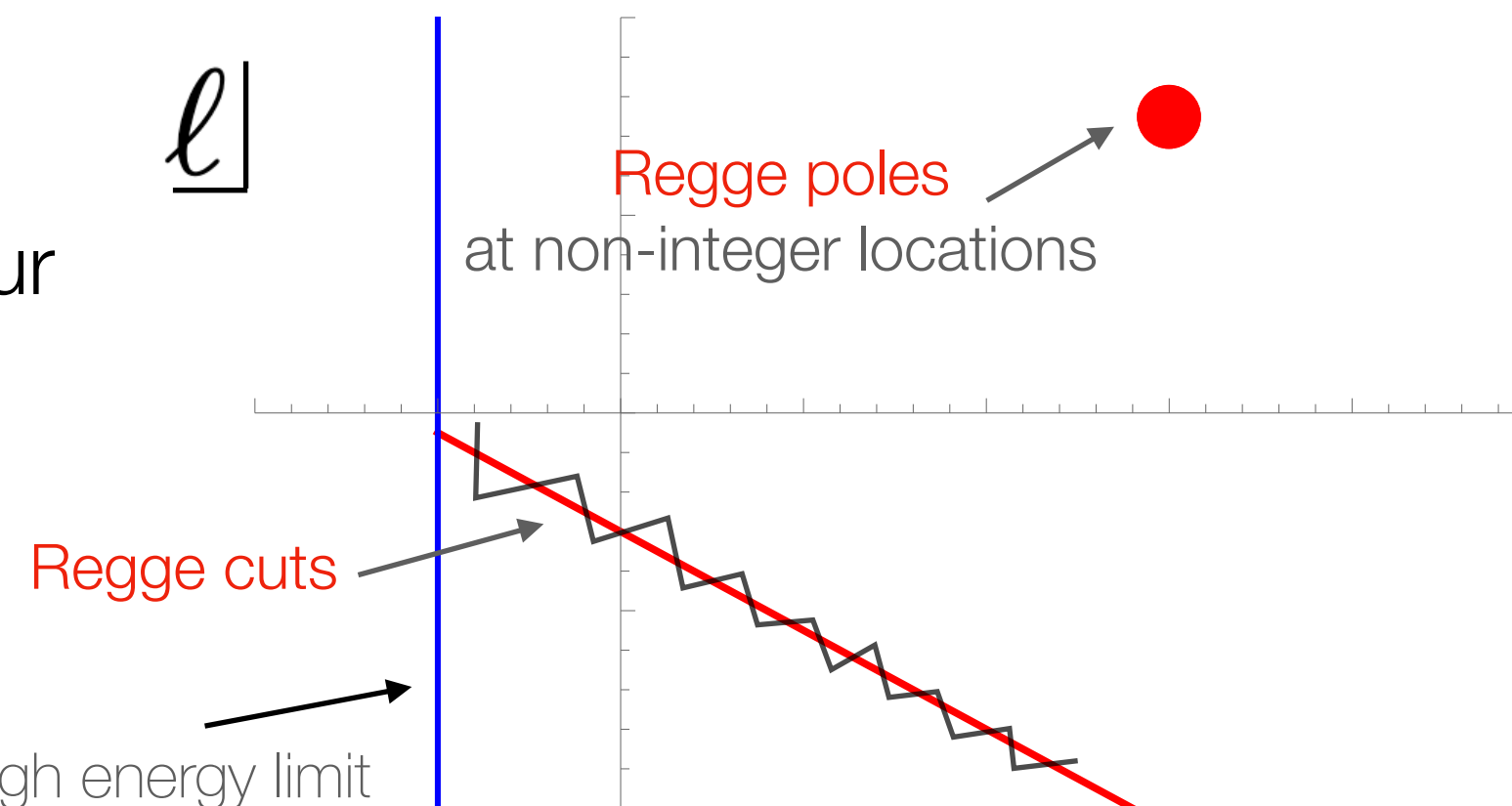
Legendre polynomials have the asymptotic behaviour

$$\lim_{s \rightarrow \infty} P_{\ell}(s) \sim s^{\ell}$$

Regge's idea was to open up the contour

Doing so we will pick up **other analytic behaviour**

This contour is subleading in high energy limit



Regge cuts arise from only **nonplanar diagrams** [Mandelstam '63]

Only from Feynman integral analysis, there is no colour (yet)

We will show how to **disentangle** cuts and poles in perturbative QCD



# The High-Energy Limit of QCD

At Leading Logarithmic Accuracy: The **Regge** trajectory

- At leading power the amplitude is

$$\mathcal{M}_{ij \rightarrow ij}^{\text{tree}} = g_s^2 \frac{2s}{t} (T_i^b)_{a_1 a_4} (T_j^b)_{a_2 a_3} \delta_{\lambda_1 \lambda_4} \delta_{\lambda_2 \lambda_3}$$

helicity conservation

colour operator notation  
valid for any representation

$$\mathbf{T}_1^c (\mathbf{T}_i^b)_{a_1 a_4} (\mathbf{T}_j^b)_{a_2 a_3} = (\mathbf{T}_1^c)_{a_1 a'_1} (\mathbf{T}_i^b)_{a'_1 a_4} (\mathbf{T}_j^b)_{a_2 a_3}$$

convenient to define the colour  
flow operators corresponding to  
Mandelstam invariants

$$\mathbf{T}_s = \mathbf{T}_1 + \mathbf{T}_2$$

$$\mathbf{T}_u = \mathbf{T}_1 + \mathbf{T}_3$$

$$\mathbf{T}_t = \mathbf{T}_1 + \mathbf{T}_4$$

colour charge in t-channel is  
the adjoint Casimir (gluon)

$$\mathbf{T}_t^2 \mathcal{M}_{ij \rightarrow ij}^{\text{tree}} = C_A \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

- Large Logarithms develop in the ratio  $s/(-t)$  at each order in  $\alpha_s$

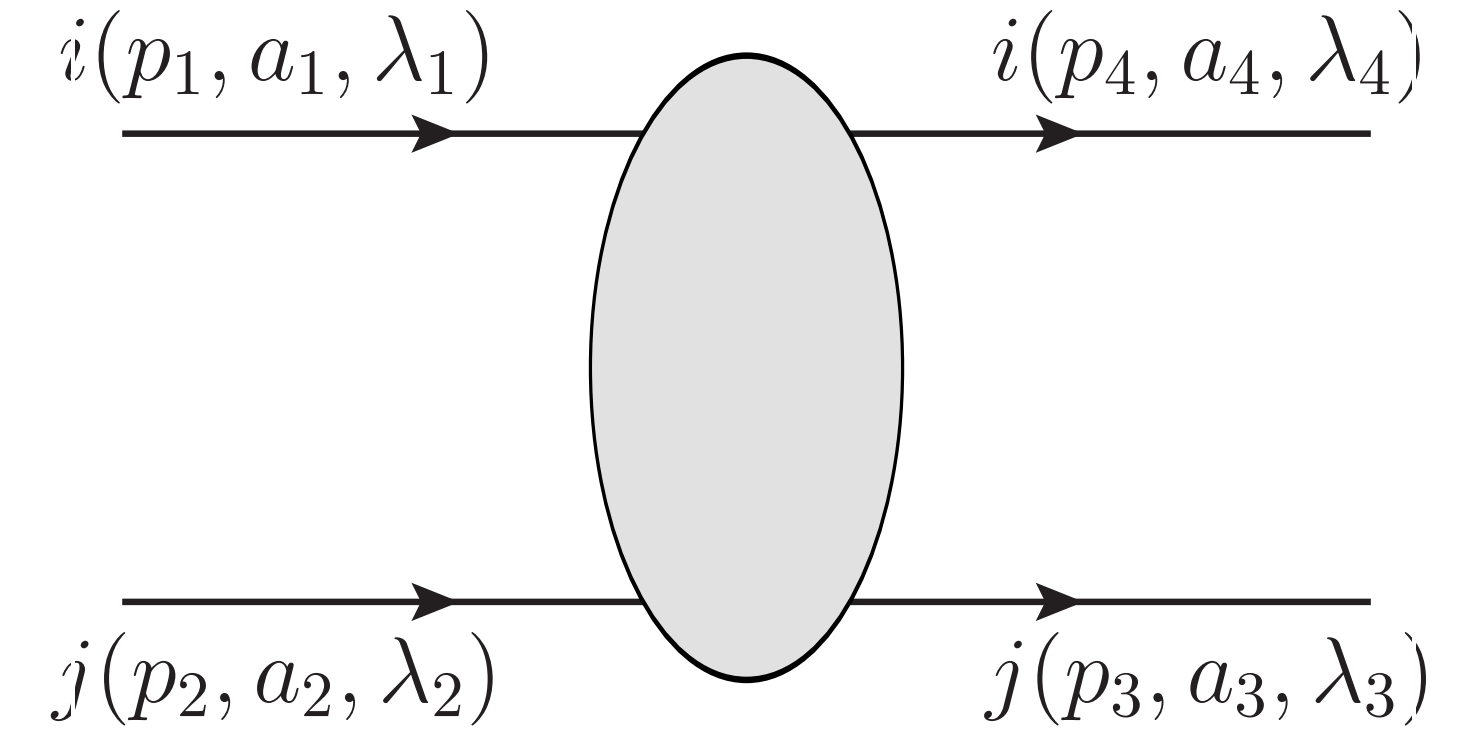
- They can be **resummed** by the famous QCD **Regge** pole

$$\mathcal{M}_{ij \rightarrow ij}^{\text{LL}} = \left( \frac{s}{-t} \right)^{C_A \alpha_g(t, \mu^2)} \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

- The exponent  $\alpha_g$  is the **Regge trajectory**. At one-loop it

$$\alpha_g^{(1)}(t, \mu^2) = \frac{r_\Gamma}{2\epsilon} \left( \frac{\mu^2}{-t} \right)^\epsilon \quad r_\Gamma = e^{\epsilon \gamma_E} \frac{\Gamma^2(1 - \epsilon) \Gamma(1 + \epsilon)}{\Gamma(1 - 2\epsilon)}$$

- This captures logarithms of the form  $\alpha_s^n \log^n \left( \frac{s}{-t} \right)$ , what about subleading logarithms?



# Odd/even amplitudes

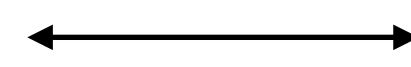
- It is convenient to exploit  $s \leftrightarrow u$  symmetry in the high-energy limit. We define even and odd amplitudes such that they have definite sign under crossing

$$\mathcal{M}^{(\pm)}(s, t) = \frac{1}{2} \left( \mathcal{M}(s, t) \pm \mathcal{M}(-s - t, t) \right)$$

- With the symmetric even logarithm

$$L \equiv \log \left( \frac{s}{-t} \right) - \frac{i\pi}{2} = \frac{1}{2} \left[ \log \left( \frac{-s - i0}{-t} \right) + \log \left( \frac{-u - i0}{-t} \right) \right]$$

Odd amplitudes



Real part

[Caron-Huot, Gardi, Vernazza '17]

Even amplitudes



Imaginary part

We will focus on the odd amplitudes

- It will be convenient to define the odd colour structure  $\mathbf{T}_{s-u}^2 = \frac{1}{2}(\mathbf{T}_s^2 - \mathbf{T}_u^2)$

As the kinematic term  $\frac{s}{t}$  is odd, the odd amplitude will have an even number of  $\mathbf{T}_{s-u}^2$

- The amplitude, written as an expansion  $\mathcal{M}_{ij \rightarrow ij}^{(\pm)} = \sum_{n=0}^{\infty} \left( \frac{\alpha_s}{\pi} \right)^n \sum_{m=0}^n L^m \mathcal{M}_{ij \rightarrow ij}^{(\pm, n, m)}$

# Next-to-Leading-Logarithms

## Two-loop Regge trajectory and impact factors

- At the next logarithmic accuracy we need to generalise the Regge pole - **Regge factorisation**

Odd amplitude  $\mathcal{M}_{ij \rightarrow ij}^{(-)} = e^{C_A \alpha_g(t) L} C_i(t) C_j(t) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$

Regge trajectory
impact factors

- Expanding to two loops we have

$$\mathcal{M}_{ij \rightarrow ij}^{(-,2)} = \left\{ \frac{1}{2} \left( C_A \alpha_g^{(1)} \right)^2 L^2 + C_A \left[ \alpha_g^{(2)} + \alpha_g^{(1)} \left( C_i^{(1)} + C_j^{(1)} \right) \right] L + C_i^{(1)} C_j^{(1)} + C_i^{(2)} + C_j^{(2)} \right\} \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

Leading logs
NLL pole parameters are the two-loop Regge trajectory and one-loop impact factors



# Infrared Divergences

## The cusp and collinear anomalous dimensions

A similar exponentiation occurs for infrared divergences of scattering amplitudes

soft anomalous dimension

Long distance singularities **factorise**

e.g. [Catani '98; Aybat, Dixon, Sterman '06; Gardi, Magnea '09; Becher, Neubert '09]

$$\mathcal{M} = \mathbf{Z} \cdot \mathcal{H}$$

finite hard function

$$\mathbf{Z} = \mathbb{P} \exp \left\{ -\frac{1}{2} \int_0^{\mu^2} \frac{d\lambda^2}{\lambda^2} \mathbf{\Gamma} \right\}$$

integrating over the running coupling generates infrared poles

**Soft anomalous dimension** in the high-energy limit is

[Del Duca, Duhr, Gardi, Magnea, White '11]

**collinear** anomalous dimensions for the external legs

$$\mathbf{\Gamma} = \frac{1}{2} \gamma_K [L \mathbf{T}_t^2 + i\pi \mathbf{T}_{s-u}^2] + \Gamma_i + \Gamma_j + \mathbf{\Delta}$$

**cusp** anomalous dimension captures the LL divergences

starts at NLL for even amplitudes and NNLL for odd amplitudes

# Comparing Regge and infrared factorisation

Regge factorisation valid to NLL

$$\mathcal{M}_{ij \rightarrow ij}^{(-)} = e^{C_A \alpha_g(t) L} C_i(t) C_j(t) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

Infrared factorisation valid to NLL

$$\begin{aligned} \mathcal{M}_{ij \rightarrow ij}^{(-)}|_{\text{poles}} &= \exp \left[ -\frac{1}{4} \int_0^{\mu^2} \frac{d\lambda^2}{\lambda^2} \gamma_K(\alpha_s(\lambda^2)) L \mathbf{T}_t^2 \right] Z_i(t) Z_j(t) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}} \\ &= e^{\mathbf{T}_t^2 K(\alpha_s) L} Z_i(t) Z_j(t) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}} \end{aligned}$$

Comparing the two exponentiations gives two obvious equalities at NLL

$$K(\alpha_s(\mu^2)) \equiv -\frac{1}{4} \int_0^{\mu^2} \frac{d\lambda^2}{\lambda^2} \gamma_K(\alpha_s(\lambda^2)) \quad \mu^2 = -t \quad Z_i(t) = \exp \left\{ -\frac{1}{2} \int_0^{\mu^2} \frac{d\lambda^2}{\lambda^2} \Gamma_i \left( \alpha_s(\lambda^2), \frac{-t}{\lambda^2} \right) \right\}$$

$$\alpha_g(t) = K + \mathcal{O}(\epsilon^0)$$

holds at two loops

$$C_{i/j}(t) = Z_{i/j}(t) D_{i/j}(t)$$

finite at one loop

[Del Duca, Falcioni, Magnea, Vernazza '14]

This identification suggests it is **eikonal**  
[Korchenskaya, Korchemsky '94, '96]

We will see later how these  
generalise at NNLL and beyond

# Next-to-Next-to-Leading-Logarithms

## Non-factorising term

- At the next logarithmic order the factorisation fails and we need to add a **non-factorising** term

$$\begin{aligned}\mathcal{M}_{ij \rightarrow ij}^{(-)} &= \mathcal{M}_{ij \rightarrow ij}^{(-), F} + \mathcal{M}_{ij \rightarrow ij}^{(-), NF} \\ &= e^{C_A \alpha_g(t) L} C_i(t) C_j(t) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}} + \mathcal{M}_{ij \rightarrow ij}^{(-), NF}\end{aligned}$$

While the factorising term is associated to **Regge poles** the non-factorising term is associated to **Regge cuts**

universal - independent of fundament/adjoint scattering particles

depends only on  $i$  or  $j$  not together

The pole term can only contain Casimirs

But the amplitude no longer carries just the colour structure of the tree-level

- To see this, let us return to **colour**

Use a basis that is **orthonormal** and one of the elements is the tree-level antisymmetric octet

$$(8 \otimes 8)_{gg \rightarrow gg} = 1 \oplus 8_s \oplus 8_a \oplus (10 \oplus \overline{10}) \oplus 27 \oplus 0$$

$$(8 \otimes 8)_{qg \rightarrow qg} = 1 \oplus 8_s \oplus 8_a$$

$$(3 \otimes \bar{3})_{qq \rightarrow qq} = 1 \oplus 8_a$$

tree-level

contributes to the odd amplitude at NNLL

the others contribute to the even amplitude



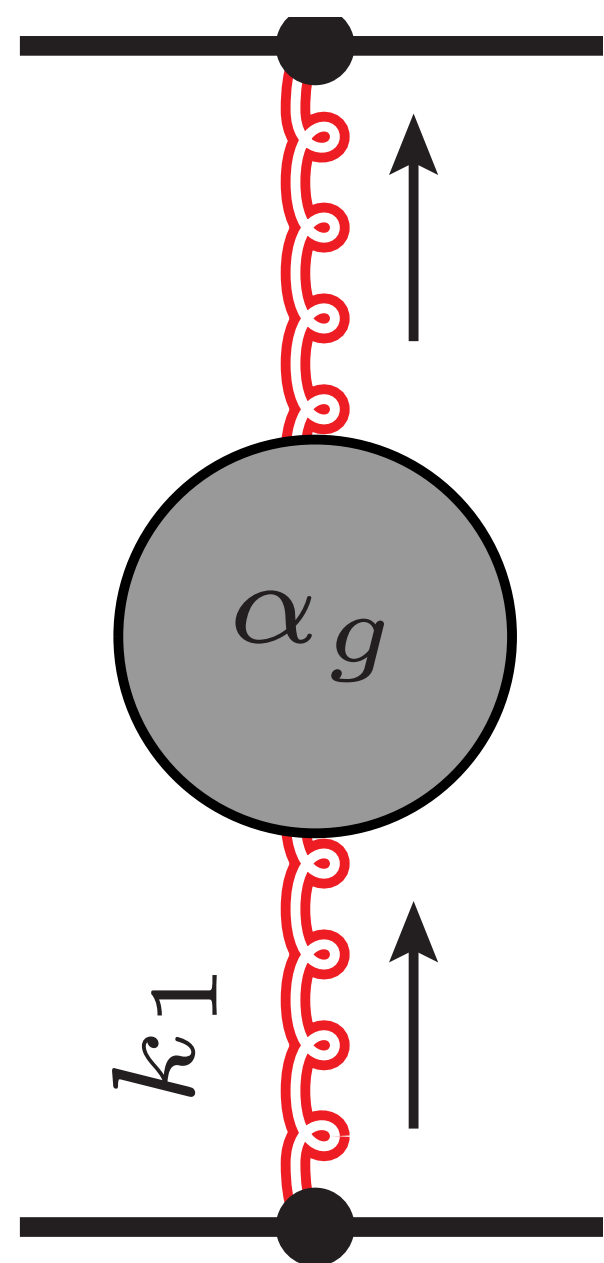
# The Scheme Dependence at NNLL

- Expanding the above two loops at NNLL we have

$$\mathcal{M}_{ij \rightarrow ij}^{(-,2,0)} = \left( \overset{\text{loop order}}{C_i^{(1)}} \overset{\text{log order}}{C_i^{(1)}} + C_i^{(2)} + \overset{\text{fixed}}{C_j^{(2)}} \right) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}} + \mathcal{M}_{ij \rightarrow ij}^{(-,2,0), \text{NF}}$$

We can shift terms that are colour proportional to the tree-level between these  
There is an **ambiguity** in the definition of the two-loop impact factors

- We can access this NF term by a computation



$$e^{C_A \alpha_g(t)L} C_i(t) C_j(t) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

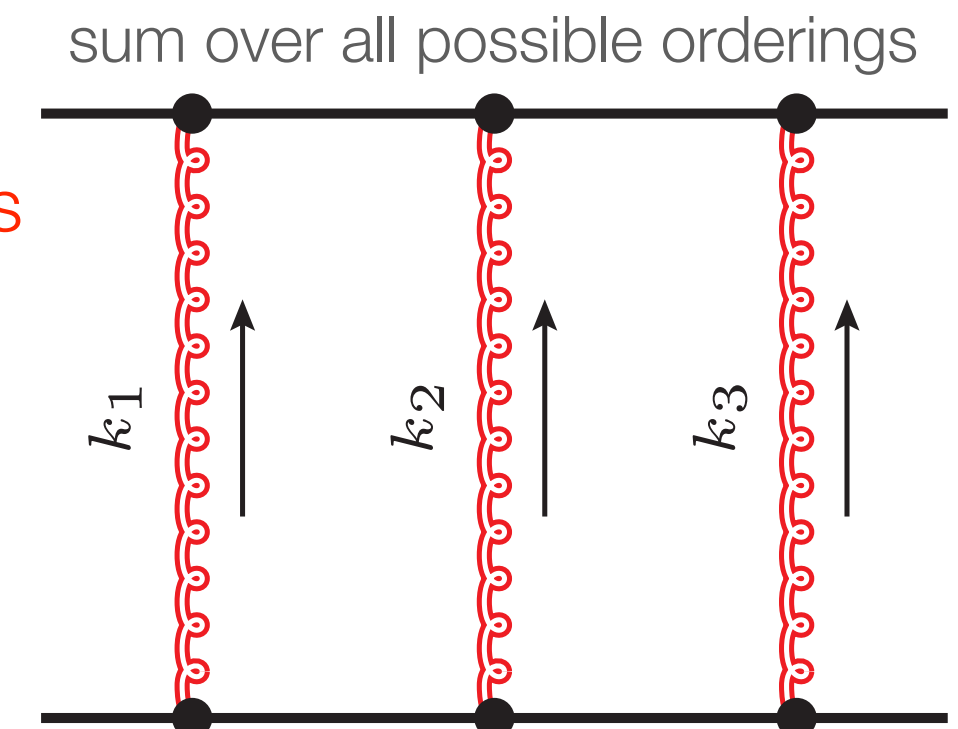
The pole term can be thought of as a single “Reggeized” gluon (**Reggeon**) in the t-channel

Two Reggeon exchange appears in the **even** amplitude. We shall not consider it.

At two loops we can start to exchange **three Reggeons**

We can calculate it based on shockwave formalism and Balitsky-JIMWLK evolution  
[Caron-Huot '13]

Overall factor of  $\pi^2 = -(i\pi)^2$  is indicative of a **cut**



Even number of  $\mathbf{T}_{s-u}^2$

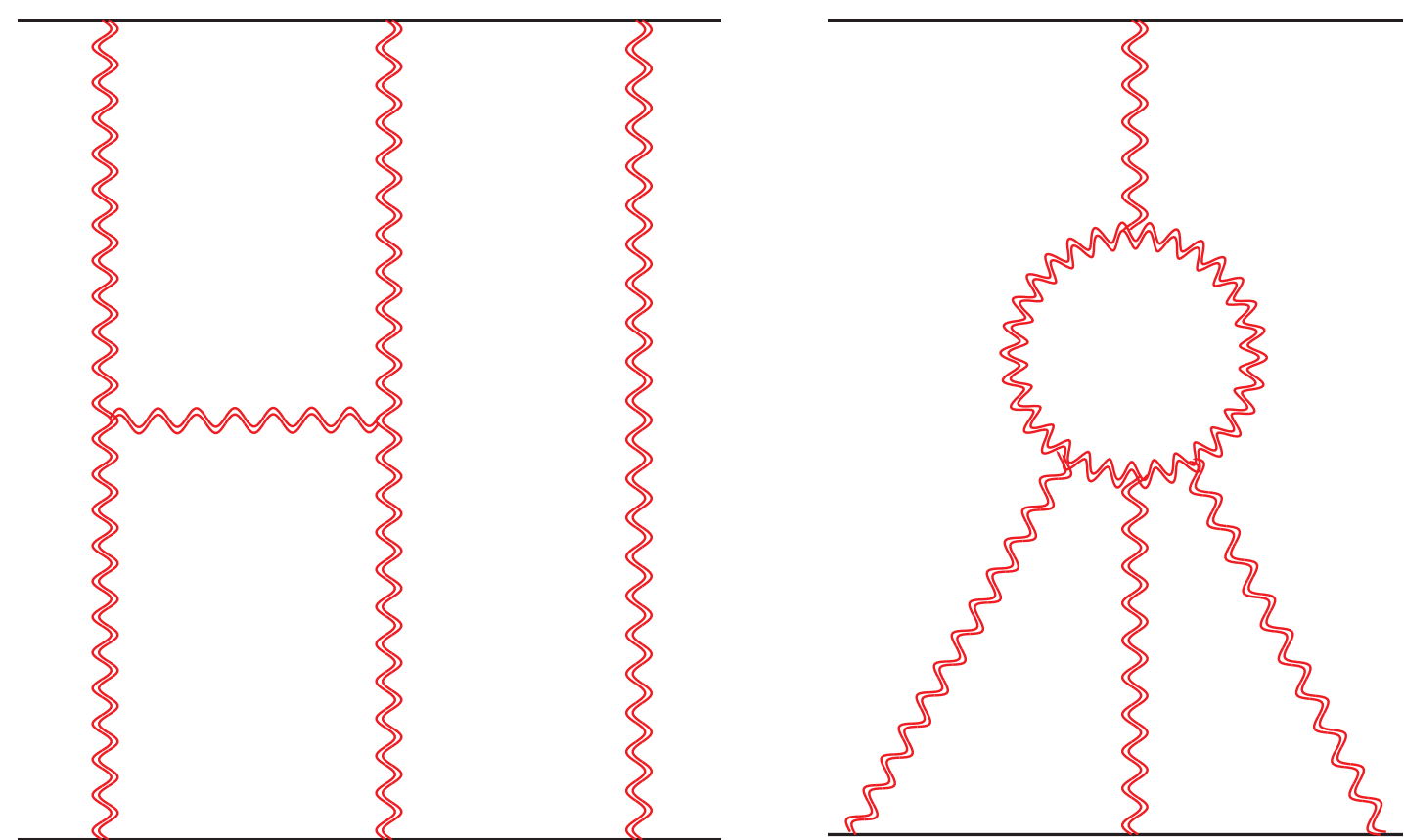
$$\mathcal{M}_{ij \rightarrow ij}^{(-,2,0), \text{MRS}} = \pi^2 \left( -\frac{1}{8\epsilon^2} + \frac{\pi^2}{48} + \frac{4}{3}\epsilon\zeta_3 + \mathcal{O}(\epsilon^2) \right) \left( (\mathbf{T}_{s-u}^2)^2 - \frac{1}{12}C_A^2 \right) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

[Caron-Huot, Gardi, Vernazza '17]

we shall call this scheme **MRS = Multiple Reggeon States**



# MRS at three and four loops



$N_c \rightarrow \infty$   
commutators are **subleading** in planar limit

only this gives **planar** contribution

$$\mathcal{M}_{ij \rightarrow ij}^{(-,3,1), \text{MRS}} = -\pi^2 r_\Gamma^3 \left[ S_A^{(3)}(\epsilon) \mathbf{T}_{s-u}^2 [\mathbf{T}_{s-u}^2, \mathbf{T}_t^2] + S_B^{(3)}(\epsilon) [\mathbf{T}_{s-u}^2, \mathbf{T}_t^2] \mathbf{T}_{s-u}^2 + S_C^{(3)}(\epsilon) C_A^3 \right] \hat{\mathcal{M}}_{\text{tree}}$$

Functions in  $\epsilon$   
Uniform transcendental weight

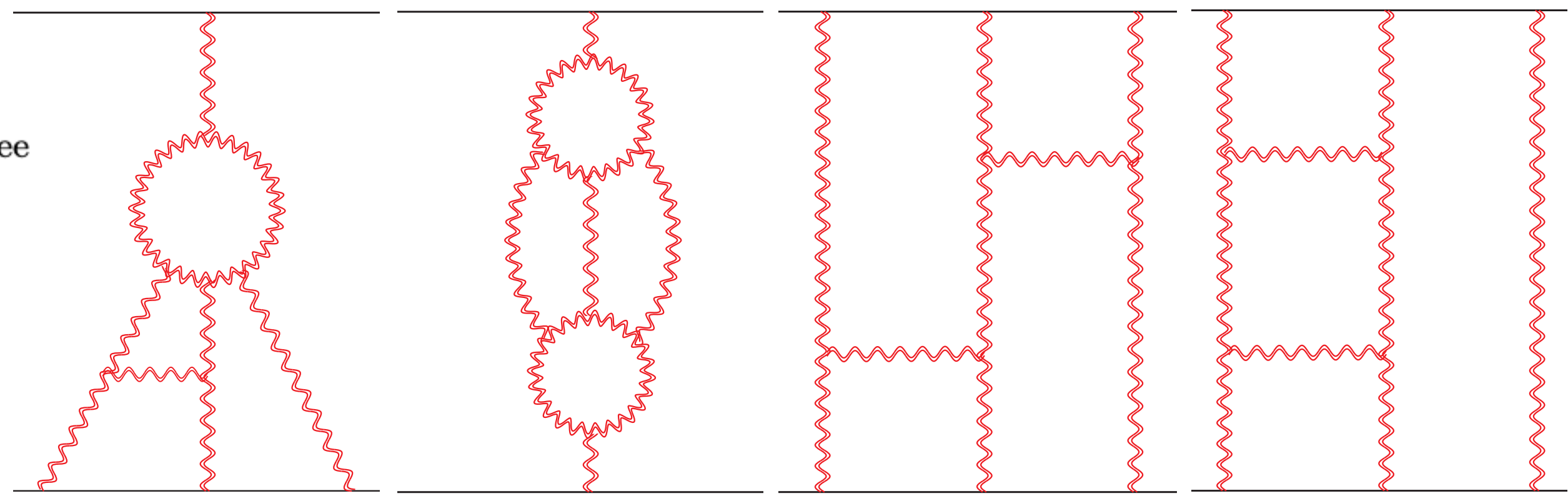
[Caron-Huot, Gardi, Vernazza '17]

Common ratios between even and odd zetas  
Define new zeta values to absorb them

They appear in other computations  
[Baikov, Chetyrkin '19; Kotikov, Teber '19]

$$\mathcal{M}_{ij \rightarrow ij}^{(-,4,2), \text{MRS}} = \frac{\pi^2 r_\Gamma^4}{2} \left[ \frac{1}{\epsilon^4} \mathbf{K}^{(4)} + \left( \frac{1}{\epsilon} \zeta_3 + \frac{3}{2} \zeta_4 \right) \mathbf{K}^{(1)} + \mathcal{O}(\epsilon) \right] \hat{\mathcal{M}}_{\text{tree}}$$

$$\begin{aligned} \mathbf{K}^{(4)} &= \frac{1}{96} [\mathbf{T}_{s-u}^2, [\mathbf{T}_{s-u}^2, \mathbf{T}_t^2]] \mathbf{T}_t^2 + \frac{7}{576} \mathbf{T}_t^2 [(\mathbf{T}_{s-u}^2)^2, \mathbf{T}_t^2] \\ &\quad - \frac{1}{192} [\mathbf{T}_{s-u}^2, \mathbf{T}_t^2] \mathbf{T}_t^2 \mathbf{T}_{s-u}^2 - \frac{5}{192} \mathbf{T}_{s-u}^2 [\mathbf{T}_{s-u}^2, \mathbf{T}_t^2] \mathbf{T}_t^2, \\ \mathbf{K}^{(1)} &= \frac{49}{48} [\mathbf{T}_{s-u}^2, [\mathbf{T}_{s-u}^2, \mathbf{T}_t^2]] \mathbf{T}_t^2 - \frac{47}{288} \mathbf{T}_t^2 [(\mathbf{T}_{s-u}^2)^2, \mathbf{T}_t^2] \\ &\quad + \frac{101}{96} [\mathbf{T}_{s-u}^2, \mathbf{T}_t^2] \mathbf{T}_t^2 \mathbf{T}_{s-u}^2 - \frac{49}{48} \mathbf{T}_{s-u}^2 [\mathbf{T}_{s-u}^2, \mathbf{T}_t^2] \mathbf{T}_t^2 + \frac{1}{24} \left( \frac{d_{AA}}{N_A} - \frac{C_A^4}{24} \right) \end{aligned}$$



← The four-loop contribution is **entirely nonplanar**

[Falcioni, Gardi, **CM**, Vernazza '20]



# Choosing a scheme

The impact factors can be found from matching to amplitudes

$$\mathcal{M}_{ij \rightarrow ij}^{(-,2,0)} = \left( C_i^{(1)} C_i^{(1)} + C_i^{(2)} + C_j^{(2)} \right) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}} + \mathcal{M}_{ij \rightarrow ij}^{(-,2,0), \text{MRS}}$$

$$\mathcal{M}_{ij \rightarrow ij}^{(-,2,0), \text{MRS}} = \pi^2 \left( -\frac{1}{8\epsilon^2} + \frac{\pi^2}{48} + \frac{4}{3}\epsilon\zeta_3 + \mathcal{O}(\epsilon^2) \right) \left( (\mathbf{T}_{s-u}^2)^2 - \frac{1}{12}C_A^2 \right) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

What is the most “physical” scheme?

But we could move any part of the term **colour proportional** to the tree-level into new impact factors

- The MRS scheme was used because it made **calculational sense**

- Evaluating the term in the **planar limit** gives  $\mathcal{M}_{ij \rightarrow ij}^{(-) \text{MRS}} \Big|_{\text{planar}} = \frac{\pi^2 r_\Gamma^2 N_c^2}{6} \left( \frac{\alpha_s}{\pi} \right)^2 \left\{ S^{(2)}(\epsilon) - \left( \frac{\alpha_s}{\pi} \right) r_\Gamma N_c L \left[ 6 S_C^{(3)}(\epsilon) - \frac{1}{2\epsilon} S^{(2)}(\epsilon) \right] \right\} \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$

- At odds with the term being identified as coming from Regge cuts in the complex angular momentum plane

- Naturally gives a **new scheme** we call the “**cut**” scheme  
[Falcioni, Gardi, Maher, **CM**, Vernazza '21, **PRL**]

$$\mathcal{M}_{ij \rightarrow ij}^{(-)} = \underbrace{\mathcal{M}_{ij \rightarrow ij}^{(-) \text{SRS}} + \mathcal{M}_{ij \rightarrow ij}^{(-) \text{MRS}} \Big|_{\text{planar}}}_{\mathcal{M}_{ij \rightarrow ij}^{(-) \text{pole}}} + \mathcal{M}_{ij \rightarrow ij}^{(-) \text{MRS}} \Big|_{\text{nonplanar}} = \mathcal{M}_{ij \rightarrow ij}^{(-) \text{pole}} + \mathcal{M}_{ij \rightarrow ij}^{(-) \text{cut}}$$

SRS = Single Reggeon State

# Introducing the cut scheme

[Falcioni, Gardi, Maher, **CM**, Vernazza '21, **PRL**]

$$\begin{aligned}\mathcal{M}_{ij \rightarrow ij}^{(-)} &= \underbrace{\mathcal{M}_{ij \rightarrow ij}^{(-) \text{SRS}} + \mathcal{M}_{ij \rightarrow ij}^{(-) \text{MRS}} \Big|_{\text{planar}}}_{\mathcal{M}_{ij \rightarrow ij}^{(-) \text{pole}}} + \mathcal{M}_{ij \rightarrow ij}^{(-) \text{MRS}} \Big|_{\text{nonplanar}} \\ &= \mathcal{M}_{ij \rightarrow ij}^{(-) \text{pole}} + \mathcal{M}_{ij \rightarrow ij}^{(-) \text{cut}}\end{aligned}$$

- We define a new pole term with a **new** Regge trajectory and new impact factors

$$\mathcal{M}_{ij \rightarrow ij}^{(-) \text{pole}} = \tilde{C}_i(t) \tilde{C}_j(t) e^{\tilde{\alpha}_g(t) C_A L} \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

- At **two loops** we compute the nonplanar piece of the MRS term.

Then keeping the **whole amplitude invariant** we have a relation between the impact factors in both schemes

$$\tilde{C}_{i/j}^{(2)} = C_{i/j}^{(2)} + N_c^2 (r_\Gamma)^2 \frac{\pi^2}{12} S^{(2)}(\epsilon)$$

- We can do the same at three loops, finding a relation between the three-loop Regge trajectories

$$\tilde{\alpha}_g^{(3)} = \alpha_g^{(3)} - (r_\Gamma)^3 N_c^2 \frac{\pi^2}{18} \left( S_A^{(3)}(\epsilon) - S_B^{(3)}(\epsilon) \right)$$

- At **four loops** (and beyond) the MRS term was **already nonplanar**

We have exhausted all possible parameters at NNLL to absorb planar contributions into the pole term

As such we know the **four-loop Regge cut** in any gauge theory! It is a **universal** result

What can we extract?



# Two-loop results

- Using the **cut scheme** we can extract two-loop Regge trajectory and impact factors

$$\mathcal{M}_{ij \rightarrow ij}^{(-,2,1), \text{pole}} = C_A \left[ \alpha_g^{(2)} + \alpha_g^{(1)} \left( C_i^{(1)} + C_i^{(1)} \right) \right] \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

Note the two-loop Regge trajectory in both schemes are equal. It is a NLL quantity

- We expand the amplitudes known to  $\mathcal{O}(\epsilon^2)$  in the high-energy limit and compare

[Ahmed, Henn, Mistlberger '19]

we define the **cusp-subtracted** Regge trajectory

$$\hat{\tilde{\alpha}}_g = \tilde{\alpha}_g - K \quad K(\alpha_s(\mu^2)) \equiv -\frac{1}{4} \int_0^{\mu^2} \frac{d\lambda^2}{\lambda^2} \gamma_K(\alpha_s(\lambda^2))$$

$$\begin{aligned} \hat{\tilde{\alpha}}_g^{(2)} = & C_A \left( \frac{101}{108} - \frac{\zeta_3}{8} \right) - \frac{7n_f}{54} + \epsilon \left[ C_A \left( \frac{607}{324} - \frac{67\zeta_2}{144} - \frac{33\zeta_3}{8} - \frac{3\zeta_4}{16} \right) + n_f \left( -\frac{41}{162} + \frac{5\zeta_2}{72} + \frac{3\zeta_3}{4} \right) \right] \\ & + \epsilon^2 \left[ C_A \left( \frac{911}{243} - \frac{101\zeta_2}{108} - \frac{1139\zeta_3}{108} - \frac{2321\zeta_4}{384} + \frac{41\zeta_5}{8} + \frac{71\zeta_2\zeta_3}{24} \right) + n_f \left( \frac{7\zeta_2}{54} + \frac{85\zeta_3}{54} + \frac{211\zeta_4}{192} - \frac{122}{243} \right) \right] + \mathcal{O}(\epsilon^3) \end{aligned}$$

- Order  $\epsilon$  and above are **new** results [Falcioni, Gardi, Maher, **CM**, Vernazza '21, **PRL**]
- Finite** at two loops, as discussed earlier
- Agrees with the planar limit of [Del Duca, Marzucca, Verbeek '21]



# Two-loop impact factors

$$\mathcal{M}_{ij \rightarrow ij}^{(-,2,0), \text{pole}} = \left( C_i^{(1)} C_i^{(1)} + \tilde{C}_i^{(2)} + \tilde{C}_j^{(2)} \right) \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

$$\mathcal{M}_{ij \rightarrow ij}^{(-,2,0) \text{ cut}} = \pi^2 (r_\Gamma)^2 S^{(2)}(\epsilon) \left[ (\mathbf{T}_{s-u}^2)^2 - \frac{C_A^2}{4} \right] \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

- We **extract** them to  $\mathcal{O}(\epsilon^2)$
- $\mathcal{O}(\epsilon)$  and above are **new** results [\[Falcioni, Gardi, Maher, CM, Vernazza '21, PRL\]](#)
- Agrees with the planar limit of [\[Del Duca, Marzucca, Verbeek '21\]](#)
- Divergences are given by the **collinear anomalous dimension**  
[\[Del Duca, Falcioni, Magnea, Vernazza '14\]](#)

There is no cut in the planar limit!  
We are considering the cut  
contribution for the first time!

# The three-loop Regge trajectory

- We can also do the same at **three loops**

$$\mathcal{M}_{ij \rightarrow ij}^{(-,3,1) \text{ pole}} = N_c \left[ \tilde{\alpha}_g^{(3)}(t) + \alpha_g^{(2)}(t) \left( C_i^{(1)} + C_j^{(1)} \right) + \alpha_g^{(1)}(t) \left( C_i^{(1)} C_j^{(1)} + \tilde{C}_i^{(2)} + \tilde{C}_j^{(2)} \right) \right] \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

- The relevant three-loop amplitudes have been computed recently [\[Caola, Chakraborty, Gambuti, von Manteuffel, Tancredi '21\]](#)

$$\begin{aligned} \hat{\tilde{\alpha}}_g^{(3)} = & C_A^2 \left( \frac{297029}{93312} - \frac{799\zeta_2}{1296} - \frac{833\zeta_3}{216} - \frac{77\zeta_4}{192} + \frac{5}{24}\zeta_2\zeta_3 + \frac{\zeta_5}{4} \right) + C_A n_f \left( \frac{103\zeta_2}{1296} + \frac{139\zeta_3}{144} - \frac{5\zeta_4}{96} - \frac{31313}{46656} \right) \\ & + C_F n_f \left( \frac{19\zeta_3}{72} + \frac{\zeta_4}{8} - \frac{1711}{3456} \right) + n_f^2 \left( \frac{29}{1458} - \frac{2\zeta_3}{27} \right) + \mathcal{O}(\epsilon), \end{aligned}$$

[\[Falcioni, Gardi, Maher, \*\*CM\*\*, Vernazza '21, \*\*PRL\*\*; see also Caola, Chakraborty, Gambuti, von Manteuffel, Tancredi '21\]](#)

also Giulio Gambuti's talk

- **Finite**, thus has poles governed by the cusp anomalous dimension

Also for  $n_f = 0$  there are no  $N_c$ -subleading corrections, maximally non-Abelian

Is it an **eikonal** quantity? Can we define it in terms of **Wilson lines**?

- Agrees with the planar limit of [\[Del Duca, Marzucca, Verbeek '21\]](#)

# Infrared Constraints at four loops

- Infrared factorisation  $\mathcal{M} = \mathbf{Z} \cdot \mathcal{H}$   $\mathbf{Z} = \mathbb{P} \exp \left\{ -\frac{1}{2} \int_0^{\mu^2} \frac{d\lambda^2}{\lambda^2} \mathbf{\Gamma} \right\}$
- Soft anomalous dimension in high-energy limit  $\mathbf{\Gamma} = \frac{1}{2} \gamma_K \left[ L \mathbf{T}_t^2 + i\pi \mathbf{T}_{s-u}^2 \right] + \mathbf{\Gamma}_i + \mathbf{\Gamma}_j + \mathbf{\Delta}$
- Knowing  $\mathcal{M}$  and demanding  $\mathcal{H}$  to be finite we can find  $\mathbf{\Delta}$

$$\mathbf{\Delta} = \sum_{n=0}^{\infty} \left( \frac{\alpha_s}{\pi} \right)^n \sum_{m=0}^n L^m \mathbf{\Delta}^{(n,m)}$$

- Explicit results in the high-energy limit, in **any gauge theory**

NLL 4-loop  
[Caron-Huot '13]  $\mathbf{\Delta}^{(4,3)} = -i\pi \frac{\zeta_3}{24} \left[ \mathbf{T}_t^2, \left[ \mathbf{T}_t^2, \mathbf{T}_{s-u}^2 \right] \right] \mathbf{T}_t^2$

NNLL 4-loop  
[Falcioni, Gardi, **CM**, Vernazza '20]  $\text{Re} \left[ \mathbf{\Delta}^{(4,2)} \right] = \zeta_2 \zeta_3 \left\{ \frac{1}{4} \mathbf{T}_t^2 \left[ \mathbf{T}_t^2, (\mathbf{T}_{s-u}^2)^2 \right] + \frac{3}{4} \left[ \mathbf{T}_{s-u}^2, \mathbf{T}_t^2 \right] \mathbf{T}_t^2 \mathbf{T}_{s-u}^2 + \frac{d_{AA}}{N_A} - \frac{C_A^4}{24} \right\}$

- There exists an **ansatz** for  $\mathbf{\Delta}^{(4)}$  in general kinematics [Becher, Neubert '19]
- **Matching** gives asymptotic limits of the four-loop functions [Falcioni, Gardi, Maher, **CM**, Vernazza '21]



# Further Constraints

[Falcioni, Gardi, Maher, **CM**, Vernazza '21]

Let us revisit the **two exponents**

soft anomalous dimension

cut-scheme Regge trajectory

$$-\frac{1}{2} \int_0^{\mu^2} \frac{d\lambda^2}{\lambda^2} \mathbf{\Gamma} \quad \Longleftrightarrow \quad C_A \tilde{\alpha}_g(t) L$$

infrared factorisation Regge pole

The poles of the Regge trajectory are given by the **cusp in the adjoint**

For the two exponents to match we need the **linear term of the soft anomalous dimension** to be the cusp

$$\left. \frac{d}{dL} \mathbf{\Gamma} \right|_{L=0} \mathcal{M}_{ij \rightarrow ij}^{\text{tree}} = \Gamma_A^{\text{cusp}} \mathcal{M}_{ij \rightarrow ij}^{\text{tree}}$$

↙ gives further constraints

The three-loop soft anomalous dimension is **bootstrapable**

[Almelid, Duhr, Gardi, McLeod, White '17]

We have found useful constraints for a potential **four-loop** bootstrap



# Conclusions

- We have shown how to define a **nonplanar Regge cut** in perturbative QCD by shifting planar terms from multiple Reggeon exchanges into the pole term

$$\begin{aligned}\mathcal{M}_{ij \rightarrow ij}^{(-)} &= \underbrace{\mathcal{M}_{ij \rightarrow ij}^{(-) \text{ SRS}} + \mathcal{M}_{ij \rightarrow ij}^{(-) \text{ MRS}} \Big|_{\text{planar}}}_{\mathcal{M}_{ij \rightarrow ij}^{(-) \text{ pole}}} + \mathcal{M}_{ij \rightarrow ij}^{(-) \text{ MRS}} \Big|_{\text{nonplanar}} \\ &= \mathcal{M}_{ij \rightarrow ij}^{(-) \text{ pole}} + \mathcal{M}_{ij \rightarrow ij}^{(-) \text{ cut}}\end{aligned}$$

nonplanar - consistent with  
Regge cut behaviour 

- The Regge trajectory in the pole term has infrared poles **governed by the cusp anomalous dimension**
- In this scheme we have found the **three-loop Regge trajectory**
- Provided constraints on the **four-loop soft anomalous dimension** relevant for a potential bootstrap
- To be explored: direct connection between **Regge cuts** in the complex angular momentum plane and **modern amplitudes**

# Thank you!

Extra Slides

# (Very) brief introduction to Regge poles and cuts

[Caron-Huot, Gardi, Vernazza '17]

Amplitude via dispersion relations

$$\mathcal{M}(s,t) = \frac{1}{\pi} \int_0^\infty \frac{d\hat{s}}{\hat{s} - s - i0} D_s(\hat{s},t) + \frac{1}{\pi} \int_0^\infty \frac{d\hat{u}}{\hat{u} + s + t - i0} D_u(\hat{u},t)$$

Mellin transformed discontinuities

$$D_s(s,t) = \frac{1}{2i} \int_{\gamma-i\infty}^{\gamma+i\infty} dj \, a_j^s(t) \left(\frac{s}{-t}\right)^j$$

$j$  is the spin of the particle in the  $t$ -channel

Even and odd  
amplitudes at leading  
power

$$\begin{aligned} \mathcal{M}^{(+)}(s,t) &= i \int_{\gamma-i\infty}^{\gamma+i\infty} \frac{dj}{\sin(\pi j)} \cos\left(\frac{\pi j}{2}\right) a_j^{(+)}(t) e^{jL}, \\ \mathcal{M}^{(-)}(s,t) &= \int_{\gamma-i\infty}^{\gamma+i\infty} \frac{dj}{\sin(\pi j)} \sin\left(\frac{\pi j}{2}\right) a_j^{(-)}(t) e^{jL}, \end{aligned}$$

Regge poles

$$a_j^{(-)}(t) \simeq \frac{1}{j-1-\alpha(t)}$$

Regge trajectory

$$\mathcal{M}^{(-)}(s,t)|_{\text{Regge pole}} \simeq \frac{\pi}{\sin \frac{\pi \alpha(t)}{2}} \frac{s}{t} e^{L \alpha(t)} + \dots$$

$j = 1 + \mathcal{O}(\alpha_s)$

Regge cuts

$$a_j^{(-)}(t) \simeq \frac{1}{(j-1-\alpha(t))^{1+\beta(t)}}$$

gluon

$$\mathcal{M}^{(-)}(s,t)|_{\text{Regge cut}} \simeq \frac{\pi}{\sin \frac{\pi \alpha(t)}{2}} \frac{s}{t} \frac{1}{\Gamma(1+\beta(t))} L^{\beta(t)} e^{L \alpha(t)} + \text{subleading logs.}$$



Specific colour structures in t-channel orthonormal basis

$$c_{qq}^{[8]} = \frac{2}{\sqrt{N_c^2 - 1}} (t^b)^{a_4}_{a_1} (t^b)^{a_3}_{a_2},$$

$$c_{qq}^{[1]} = \frac{1}{N_c} \delta^{a_4}_{a_1} \delta^{a_3}_{a_2},$$

$$c_{gg}^{[8_a]} = \frac{1}{N_c} \frac{1}{\sqrt{N_c^2 - 1}} f^{a_1 a_4 b} f^{a_2 a_3}_b,$$

$$c_{gg}^{[10+1\bar{0}]} = \sqrt{\frac{2}{(N_c^2 - 4)(N_c^2 - 1)}} \left[ \frac{1}{2} (\delta^{a_1}_{a_2} \delta^{a_3}_{a_4} - \delta^{a_3}_{a_1} \delta^{a_4}_{a_2}) - \frac{1}{N_c} f^{a_1 a_4 b} f^{a_2 a_3}_b \right],$$

$$c_{qg}^{[8_a]} = \sqrt{\frac{2}{N_c(N_c^2 - 1)}} (t^b)^{a_4}_{a_1} i f^{a_2 a_3 b}.$$



# Formulating highly energetic partons as Wilson lines

[Caron-Huot '13; Caron-Huot, Gardi, Vernazza '17]

“eikonal approximation”

$$U(z_\perp) = \mathbf{P} \exp \left[ i g_s \mathbf{T}^a \int_{-\infty}^{\infty} dx^+ A_+^a(x^+, x^- = 0, z_\perp) \right]$$

Fourier conjugate of  $t$

regulate rapidity divergences by tilting Wilson-line off the light cone

$$\eta = \frac{1}{2} \log \frac{p_+}{p_-}$$

$$T_{i,L}^a = [\mathbf{T}^a U(z_i)] \frac{\delta}{\delta U(z_i)}$$

$$T_{i,R}^a = [U(z_i) \mathbf{T}^a] \frac{\delta}{\delta U(z_i)}$$

our parton is a collection of such Wilson lines

evolves according to Balitsky-JIMWLK

$$\frac{d}{d\eta} |\psi_i\rangle = -H |\psi_i\rangle \quad H = \frac{\alpha_s}{2\pi^2} \int d^d z_i d^d z_j d^d z_0 \frac{z_{0i} \cdot z_{0j}}{(z_{0i}^2 z_{0j}^2)^{1-\epsilon}} \left\{ T_{i,L}^a T_{j,L}^a + T_{i,R}^a T_{j,R}^a - U_{\text{ad}}^{ab}(z_0) (T_{i,L}^a T_{j,R}^b + T_{j,L}^a T_{i,R}^b) \right\}$$

The amplitude is then written as

$$\mathcal{M}_{ij \rightarrow ij} \sim \langle \psi_j | e^{-H L} | \psi_i \rangle$$

Target

Projectile

Target and projectile separated by some

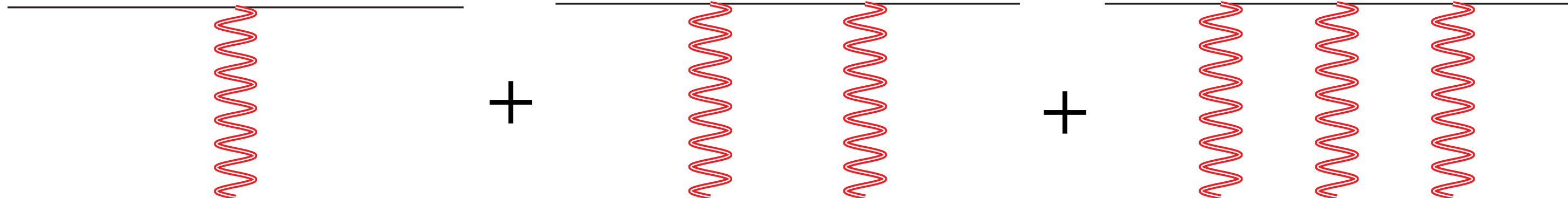
$$z_\perp^2 \leftrightarrow t$$

Evolve so that they are at equal rapidity

Expand Wilson line in Reggeons

$$U = \exp [i g_s \mathbf{T}^a W^a] \sim$$

Reggeon field



Only odd/even number of Reggeons contribute to the odd/even amplitude

In this framework the NLL divergences have been resummed and finite parts known to very high loop order

[Caron-Huot, Gardi, Reichel, Vernazza '17, '20]

$$\hat{\mathcal{M}}_{\text{NLL}}^{(+)}|_s = \frac{i\pi}{L(C_A - \mathbf{T}_t^2)} \left( 1 - R(\epsilon) \frac{C_A}{C_A - \mathbf{T}_t^2} \right)^{-1} \times \left[ \exp \left\{ \frac{B_0(\epsilon)}{2\epsilon} \frac{\alpha_s}{\pi} L(C_A - \mathbf{T}_t^2) \right\} - 1 \right] \mathbf{T}_{s-u}^2 \mathcal{M}^{(\text{tree})} + \mathcal{O}(\epsilon^0).$$

Can we achieve the same at NNLL?

# Explicit momentum-space Hamiltonians

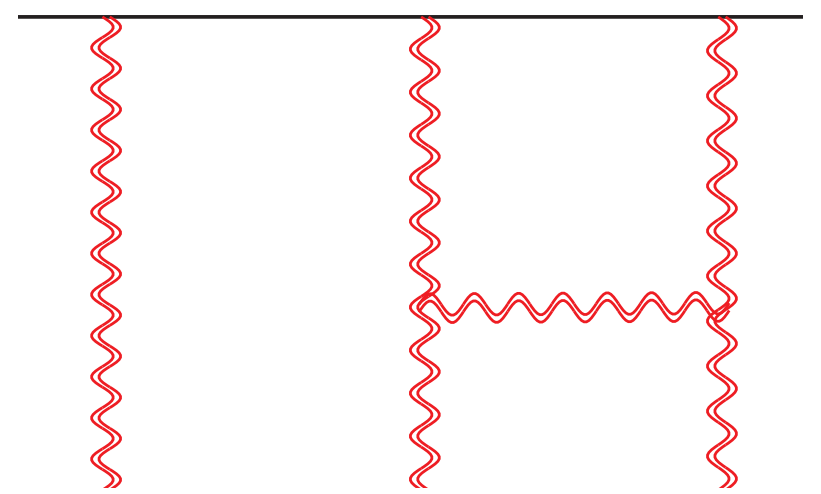
The diagonal transitions are

dresses one Reggeon  
with the trajectory

$$H_{k \rightarrow k} = - \int d^d p C_A \alpha_g(p) W^a(p) \frac{\delta}{\delta W^a(p)}$$

$$+ \alpha_s \int d^d q d^d p_1 d^d p_2 H_{22}(q; p_1, p_2) W^x(p_1 + q) W^y(p_2 - q) (F^x F^y)^{ab} \frac{\delta}{\delta W^a(p_1)} \frac{\delta}{\delta W^b(p_1)}$$

adds a *rung* between two Reggeons



Source of the difficulty at NNLL.

Three Reggeons spoil the symmetry between colour and kinematics, which is there for two Reggeons (NLL).

with kernel  $H_{22}(q; p_1, p_2) = \frac{(p_1 + p_2)^2}{p_1^2 p_2^2} - \frac{(p_1 + q)^2}{p_1^2 q^2} - \frac{(p_2 - q)^2}{q^2 p_2^2}$

The off-diagonal transitions are

$$H_{1 \rightarrow 3} = \alpha_s^2 \int d^d p_1 d^d p_2 d^d p \operatorname{tr} [F^a F^b F^c F^d] W^b(p_1) W^c(p_2) W^d(p_3) H_{13}(p_1, p_2, p_3) \frac{\delta}{\delta W^a(p)}$$

with kernel  $H_{13}(p_1, p_2, p_3) = \frac{r_\Gamma}{3\epsilon} \left[ \left( \frac{\mu^2}{(p_1 + p_2 + p_3)^2} \right)^\epsilon + \left( \frac{\mu^2}{p_2^2} \right)^\epsilon - \left( \frac{\mu^2}{(p_1 + p_2)^2} \right)^\epsilon - \left( \frac{\mu^2}{(p_2 + p_3)^2} \right)^\epsilon \right]$

# The odd NNLL amplitude

$$\mathcal{M}_{ij \rightarrow ij} \sim \langle \psi_j | e^{-HL} | \psi_i \rangle$$

Expand in terms of the coupling and logarithms

$$|\psi_i\rangle = \sum_{k=1}^{\infty} g_s^k |\psi_{i,k}\rangle$$

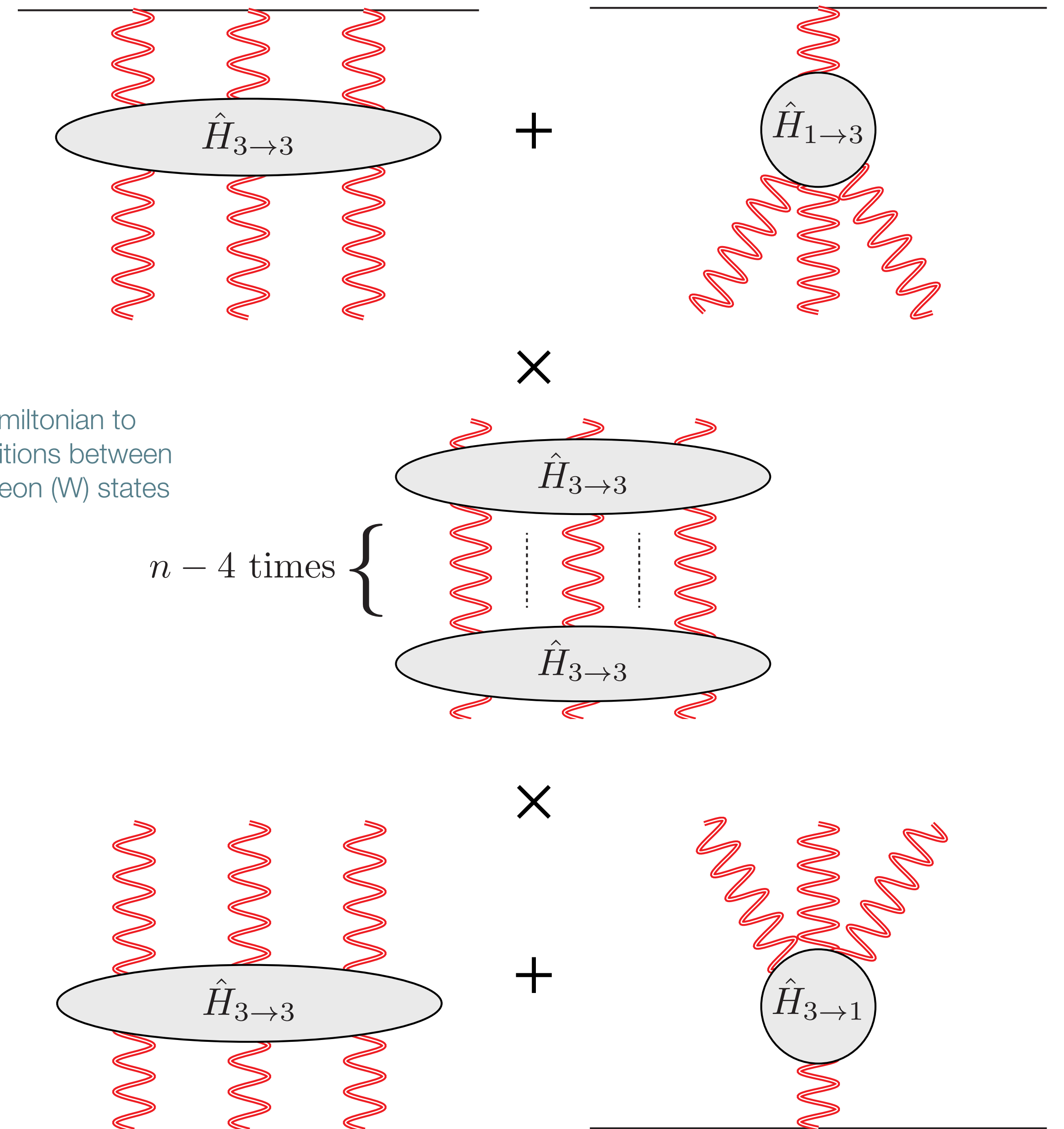
$$\hat{H}_{m \rightarrow n} \sim g_s^{|m-n|}$$

Expand Hamiltonian to describe transitions between different Reggeon (W) states

We get the all-order expression for the n-loop amplitude

$$\begin{aligned} \hat{\mathcal{M}}_{ij \rightarrow ij}^{(-), \text{NNLL}} &\sim \left( \frac{\alpha_s}{\pi} \right)^n \left( \langle \psi_{j,3} | \hat{H}_{3 \rightarrow 3} + \langle \psi_{j,1} | \hat{H}_{3 \rightarrow 1} \right) \\ &\quad \times \hat{H}_{3 \rightarrow 3}^{n-4} \\ &\quad \times \left( \hat{H}_{3 \rightarrow 3} | \psi_{j,3} \rangle + \hat{H}_{1 \rightarrow 3} | \psi_{j,1} \rangle \right) \\ &\quad + \text{explicit two-loop information} \end{aligned}$$

Only require Leading Order Balitsky-JIMWLK Hamiltonian and results from two-loop amplitudes (impact factors)





# NLL pole parameters

## Two-loop Regge trajectory

[Fadin, Kotsky, Fiore '95; Fadin, Fiore, Kotsky '96;  
Fadin, Fiore, Quartarolo '96; Blumlein, Ravindran, van Neerven '98]

$$\alpha_g^{(2)} = -\frac{b_0}{16\epsilon^2} + \frac{1}{\epsilon} \left[ C_A \left( \frac{67}{144} - \frac{\zeta_2}{8} \right) - \frac{5n_f}{72} \right] + C_A \left( \frac{101}{108} - \frac{\zeta_3}{8} \right) - \frac{7n_f}{54} + \mathcal{O}(\epsilon)$$

## One-loop impact factors

[Fadin, Fiore '92; Fadin, Lipatov '93; Fadin, Fiore, Quartarolo '94;  
Lipatov '97; Del Duca, Schmidt '98; Bern, Del Duca, Schmidt '98]

$$C_q^{(1)} = -\frac{C_F}{2\epsilon^2} - \frac{3C_F}{4\epsilon} + C_A \left( \frac{85}{72} + \frac{3\zeta_2}{4} \right) + C_F \left( \frac{\zeta_2}{4} - 2 \right) - \frac{5n_f}{36} + \mathcal{O}(\epsilon)$$

$$C_g^{(1)} = -\frac{C_A}{2\epsilon^2} - \frac{b_0}{4\epsilon} + C_A \left( \zeta_2 - \frac{67}{72} \right) + \frac{5n_f}{36} + \mathcal{O}(\epsilon)$$